5.2 Complex Numbers

Proposition 5.2.1 (Existence of Complex Numbers)
Let K be a field for which -1 has no square root (and in particular $\operatorname{char}(K) \neq 2$). Then

$$K[i] \coloneqq K[X]/(X^2+1)$$

is a field with a K-basis $\{1, i\}$ such that $i^2 = -1$. Multiplication in $K[i]$ is defined as follows

$$(a+bi)(c+di) = (ac-bd) + (bc+ad)i$$

and the inverse is given by

$$(a+bi)^{-1} = \frac{a-bi}{a^2+b^2}$$

The field extension $K(i)/K$ is Galois of degree two with

$$\operatorname{Gal}(K(i)/K) = \{1, \bar{\cdot}\}$$

*where $\bar{\cdot}$ is the **complex conjugation** automorphism given by*

$$\overline{a+bi} = a-bi$$

In particular we see that $\bar{\cdot}$ satisfies the following properties

$$\overline{\alpha+\beta} = \overline{\alpha} + \overline{\beta}$$
$$\overline{\alpha\beta} = \overline{\alpha}\,\overline{\beta}$$

Proof. The polynomial X^2+1 is of degree 2 so it is irreducible if and only if has no root. Therefore the quotient ring is a field by (3.18.40). $\mathbb{R}$ satisfies this property by (5.1.5), so $\mathbb{C}$ is a field.

Observe that as polynomials in $K[X]$

$$(a+bX)(c+dX) = ac + (bc+ad)X + bdX^2 = (ac-bd) + (bc+ad)X + bd(X^2+1)$$

from which the multiplication identity follows immediately.

For the inverse we claim that $a \neq 0$ or $b \neq 0 \implies a^2+b^2 \neq 0$. For otherwise either a/b or b/a would be a square root of -1. Therefore the expression for the inverse is well-defined and it may be verified directly to be an inverse.

The complex conjugation operator exists by (3.18.49). Therefore $\operatorname{Aut}(K(i)/K)$ has order at least $[K(i):K] = 2$, and we deduce $K(i)/K$ is Galois (3.18.107). ∎

Definition 5.2.2 (Complex Numbers)
*Define the **complex numbers** $\mathbb{C}$ to be the field $\mathbb{R}[i]$ and identify $\mathbb{R}$ with the subfield $\{a+0i \mid a \in \mathbb{R}\}$.*

*Define the **absolute value** on $\mathbb{C}$ as follows*

$$|a+bi| \coloneqq \sqrt{a^2+b^2} \in \mathbb{R}_{\geq 0}$$

*Define the **real part** and **imaginary part** as follows*

$$\operatorname{Re}(a+bi) = a$$
$$\operatorname{Im}(a+bi) = b$$

Observe that

$$\alpha + \overline{\alpha} = 2\operatorname{Re}(\alpha)$$

Proposition 5.2.3 (Complex Absolute Value)
Let $\alpha, \beta \in \mathbb{C}$ be complex numbers. The following properties hold

$$|\alpha| = |\overline{\alpha}|$$
$$|\alpha|^2 = \alpha\overline{\alpha}$$
$$|\alpha\beta| = |\alpha||\beta|$$
$$|\alpha+\beta| \leq |\alpha| + |\beta|$$

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Furthermore this agrees with the usual notion of absolute value on $\mathbb{R}$.

Proof. The first two relations follow by direct calculation. Then

$$|\alpha\beta|^2 = \alpha\beta\overline{\alpha\beta} = \alpha\overline{\alpha}\beta\overline{\beta} = |\alpha|^2|\beta|^2$$

and the third relation follows by taking square roots. Further

$$\begin{aligned}|\alpha+\beta|^2 &= (\alpha+\beta)\left(\overline{\alpha}+\overline{\beta}\right) = \alpha\overline{\alpha} + \left(\alpha\overline{\beta}+\overline{\alpha}\beta\right) + \beta\overline{\beta} \\ &= |\alpha|^2 + 2\,\mathrm{Re}(\alpha\beta) + |\beta|^2 \\ &\leq |\alpha|^2 + 2|\alpha\beta| + |\beta|^2 \\ &= (|\alpha| + |\beta|)^2\end{aligned}$$

and the result follows by taking square roots (and this being an increasing function). ■

Proposition 5.2.4 (Complex Numbers are Complete)
$\mathbb{C}$ *is complete as a normed vector space.*

Proof. Let (α_n) be a cauchy sequence in $\mathbb{C}$, and consider the decomposition into real and imaginary parts

$$\alpha_n = a_n + b_n i$$

Then

$$|a_n| \leq |\alpha_n|$$

$$|b_n| \leq |\alpha_n|$$

Therefore we see both (a_n) and (b_n) are cauchy sequences of reals, which therefore have limits a and b respectively. We may verify directly that $\alpha_n \to \alpha$ where $\alpha := a + bi$. ■

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